

Lightweight Cross-Domain Authentication Scheme for Securing Wireless IoT Devices Using Backscatter Communication

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Abstract—Cross-domain collaboration under wireless communication scenarios has gained traction in Internet of Things (IoT) applications. Authentication is essential for ensuring the security of wireless IoT devices (IoTDs). However, the existing cryptographic and physical-layer authentication schemes are unappealing in cross-domain scenarios due to the presence of resource-limited IoTDs, privacy concerns of cross-domain sharing, and the negative effect of malicious attackers. This article proposes FedScatter, a lightweight cross-domain authentication scheme for securing wireless IoTDs using backscatter communication. First, device identity signatures are constructed by harnessing passive signal features generated from feather-light backscatter tags, incurring negligible overhead. Subsequently, a federated learning model is designed to aggregate device identity information across domains while respecting device heterogeneity and data privacy. A novel parameter aggregation algorithm is proposed to bolster authentication resilience and against malicious attacks to avoid model pollution by powerful attackers with substantial hardware resources. A FedScatter prototype is implemented and evaluated, demonstrating significant improvements over state-of-the-art works in both true positive rate and false positive rate under various attacks.

Index Terms—Backscatter, federated learning (FL), Internet of Things (IoT) security, lightweight cross-domain authentication.

I. INTRODUCTION

THE CONTINUOUS advancement of wireless communication technologies drives the proliferation of efficient, intelligent, and convenient Internet of Things (IoT) applications in our daily life [1], such as the smart logistics, remote healthcare, and intelligence industrial control [2], [3], [4]. Collaboration and data sharing across domains under wireless communication scenario are vital in unlocking the full potential of IoT devices (IoTDs). For instance, the integration of healthcare IoTDs and third-party drones is considered integral in enabling precise delivery of medical supplies [5]. However, wireless IoTDs are vulnerable to various malicious attacks because of the nature of broadcast signals and open wireless channels under wireless communication scenario [6].

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An essential method to defend against the above attacks is device authentication [7], which confirms the legitimacy of the devices before allowing further interaction. Conventional schemes mainly rely on cryptographic authentication (CA) [8] that may require complex computation, failing to be applied in resource-limited wireless IoTDs [9]. Though some existing works realize the lightweight CA schemes [10], [11], the devices in these works still need extra computing and communication overhead. Considering the characteristics of wireless IoTDs, some works propose to conduct authentication by exploiting wireless signal features, such as received signal strength (RSS), channel state information (CSI), and Angle of Arrival (AoA) [12], [13], [14]. Physical-layer authentication (PLA) provides an information-theory security [15]. It does not require computation-intensive cryptography operations and, thus, is more suitable for resource-limited IoT applications.

Nevertheless, implementing existing PLA schemes for cross-domain applications in wireless communication scenarios poses significant challenges. For example, Xiong and Jamieson [16] used bulky multiantennas to construct precise signatures. Zhang et al. [14] realized an AoA-based authentication scheme using a large antenna array. These schemes are impractical for certain wireless IoTDs with a single antenna and limited power regarding cross-domain authentication. Moreover, the presence of heterogeneous wireless IoTDs (HW-IoTDs) and dynamic variation of authenticate objects pose an obstacle to selecting a representative set of physical-layer features for cross-domain authentication of devices. Furthermore, Fang et al. [17] combined the physical-layer features of the communication signal of wireless IoTD with a machine learning (ML) approach, enhancing the adaptability of wireless device authentication to complex and dynamic wireless environments. Oligieri et al. [18] proved that remote wireless communication can adopt an AI-based PLA scheme. These approaches necessitate the collection of identity data from all wireless IoTDs and will bring privacy concerns [19]. Within the PLA scheme, the identity features of wireless IoTDs may include massive amounts of privacy-sensitive information pertaining to wireless IoTDs, such as transmitter power, antenna distribution, and the motion state of devices. What is more, the extensive leak of device privacy-sensitive information serves as a shortcut for attackers to carry out forged identities and counterfeit legitimate devices. Fang et al. [20] proposed a malicious attack method in which a powerful attacker can create fake wireless channel characteristics and inject them

into a target receiver, invalidating the authentication schemes. However, few existing schemes offer sufficient defense against malicious attacks. None of the above works can support secure authentication between cross-domain wireless IoTs. Therefore, the lack of a practical cross-domain authentication for securing wireless IoTs under wireless communication scenarios inspires the following aspiration:

Can we design a lightweight cross-domain authentication scheme that adopt to wireless communication scenarios from attacks against the authentication scheme?

We propose FedScatter, a novel lightweight cross-domain scheme that appeals to cross-domain authentication and protects device security. In FedScatter, an HW-IoTD “passively” creates an identity signature by deploying several lightweight backscatter tags and cross-domain shares these signatures securely between HW-IoTDs. Then, an HW-IoTD can “sign” its identity signature on its active communication signal to prove its legitimacy. In particular, our first core insight is that backscatter communications can create lightweight and effective identity signatures via the uniqueness and certainty of backscatter tags’ spatial distribution. When an IoT active communicate with another device, its backscatter tags can absorb the signal energy, reflect several ambient radio signals, and generate a specific multipath feature that can be extracted for identity validation. The second core insight is that federated learning (FL) can protect sensitive identity and alleviate the influence of heterogeneity for cross-domain authentication. FedScatter uses FL to enable the distributed HW-IoTDs to share the identity “knowledge” in a privacy-preserving way. It also reduces the cost of primary data collection and the need for computing resources compared with centralized ML. More importantly, FL avoids the powerful attacker with enough prior knowledge and hardware resources to complete the attack.

Challenges: To design a lightweight cross-domain authentication for HW-IoTDs, we must address the following two challenges.

1) *How can a common identity signature be constructed for wireless IoTs in cross-domain authentication?*

Considering the diverse applications of IoT, some HW-IoTD may be single antenna, hardware resources-limited, and mobility. On the one hand, the single antenna and hardware resources-limited devices cannot “actively” extract ample signal characteristics, such as AoA and CSI, from the transmitter to construct the identity signatures of the devices. On the other hand, spatial attributes of the moving devices are continuously changing, making an extra puzzle for PLA to ideally confirm the device’s identity. To overcome this challenge, we design a novel backscatter-based identity signature mechanism that reflects the device’s identity by passively signing the active communication signals. The raw data of the identity signature is a 1-D signal sequential, which resources-limited devices, such as single-antenna devices, can ideally receive. Meanwhile, we apply continuous authentication that exploits the backscatter signal features of two consecutive rounds of communication and their cosine similarity as a complete identity signature vector for wireless IoTs. In this way, wireless IoTs can trace the movement of authenticated devices.

2) *How can we effectively share the identity signature information of HW-IoTDs between different domains to improve authentication performance?*

Due to the resources differentiation of heterogeneous IoTs in diverse domains, the size of identity signatures of devices may be inconsistent, which leads to a degradation of authentication performance. In order to deal with this challenge, we use FL to train a universal backscatter-based authentication model for HW-IoTDs. In particular, we consider all IoTs local to a domain as an IoT cluster. Then, we let the IoTs in an IoT cluster with the same number of backscatter tags, and the numbers of backscatter tags in different IoT clusters could be diverse. Each IoT cluster collects its identity signatures and trains a local authentication model. Then, all IoT clusters send their local model parameters to an aggregation server (AS). The AS updates global model parameters and returns to all IoT clusters. The IoT clusters update their model by global parameters and gain the final authentication model. Meanwhile, we customize an aggregation algorithm based on backscatter propagation characteristics for effectively sharing identity signature information. In addition, using FL to realize a privacy-preserving method for HW-IoTDs protects the identity signatures locally.

Summary of Results: On the one hand, we prove FedScatter is secure against various attacks. On the other hand, we implement FedScatter with several backscatter tags, universal software radio peripheral (USRP), and a server. We evaluate our authentication scheme under semi-honest attacks and powerful attacks, where the evaluation results show that FedScatter can achieve an accuracy of 97.66% for device authentication. Overall, the true positive rate (TPR) of FedScatter is improved by the ratio of 33.88% compared to the baseline scheme under the semi-honest attacks, and the false positive rate (FPR) of FedScatter is decreased by the ratio of 30.51% compared to the baseline scheme under the powerful attacks.

Specifically, the main contributions of our work in this article are summarized as follows.

- 1) We propose FedScatter, a lightweight cross-domain authentication scheme for HW-IoTDs. It combines backscatter communication and FL to enhance authentication security and privacy against malicious attacks in a wireless network.
- 2) We construct the identity signatures for HW-IoTDs via the uniqueness and certainty of backscatter tags’ spatial distribution and create an extra similarity feature to ensure that the authentication model can accurately distinguish the illegitimate HW-IoTDs and mobile HW-IoTDs. Then, we customize a novel FL aggregation algorithm to improve the resilience to defend against powerful attackers in FedScatter.
- 3) We implement FedScatter by multiple USRP devices equipped with backscatter tags, and then we evaluate the authentication performance under various attacks. The results show that FedScatter has satisfactory accuracy for authentication in HW-IoTDs.

The remainder of this article is organized as follows. In Section II, we review the related IoT authentication schemes. In Section III, we introduce the system model of FedScatter.

TABLE I
COMPARISON OF FEDSCATTER WITH OTHER EXISTING AUTHENTICATION SCHEMES

Schemes	Methodology	Spoof Attack	Sybil Attack	Relay Attack	Man-in-the-Middle Attack	Powerful Attack Resilience	Mutual Authentication	Continuous Authentication	Resource Protection	Cross-domain Device Friendly	Mobility Support	Privacy Protection
BCAuth [9]	PLA	✓	×	✓	✓	×	✓	✓	✓	×	✓	×
ShieldScatter [21]	PLA	✓	✓	×	✓	×	✓	×	✓	×	×	×
ScatterID [22]	PLA	✓	×	×	✓	×	✓	✓	✓	×	✓	×
LCDMA [10]	CA	✓	×	✓	✓	×	✓	×	✓	✓	✓	✓
AP-CDE [11]	CA	✓	×	✓	✓	✓	✓	×	✓	✓	✓	✓
X. Li et al. [23]	PLA	✓	×	✓	✓	✓	✓	✓	✓	✓	✓	×
He Fang et al. [17]	PLA	✓	×	✓	✓	✓	✓	×	×	×	✓	×
M. Abdrabou et al. [24]	PLA	✓	×	✓	✓	×	✓	×	×	✓	✓	×
A. Yazdinejad et al. [25]	PLA	✓	×	✓	✓	✓	✓	×	×	×	✓	✓
X. Qiu et al. [26]	PLA	✓	×	✓	✓	✓	×	×	×	×	×	×
FedScatter	PLA	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

In Section IV, we present the detailed design of FedScatter, including device identity signature design based on backscatter communication, universal authentication model construction, and device authentication flow. In Section V, we formally analyze the security of FedScatter. In Section VI, we implement the FedScatter and evaluate its performance based on practical signal collection. Finally, we conclude this article and indicate future work.

II. RELATED WORK

This section reviews the existing works of authentication scheme, including PLA and ML-based authentication. We compare recent related authentication schemes with FedScatter in Table I.

A. Physical-Layer Authentication

A series of studies focus on device authentication by constructing a physical-layer identity signature. Xiong and Jamieson [16] proposed SecureArray, a system using AoA to construct sensitive signatures to defend in depth against active attacks. Implementing SecureArray needs more than eight antennas and complex initiating calibration, which are unacceptable for resource-limited IoTs. Liu et al. [27] used CSI to conduct a fine-grained user authentication framework. The proposed framework can build a unique profile under the spoofing attack and achieves high accuracy. Wang et al. [28] proposed Hu-Fu, the first authentication protocol for radio frequency identification (RFID) against tag counterfeiting, signal replay, signal compensation, and brute-force feature reply.

Backscatter communication-based authentication is a novel and lightweight PLA scheme. Wang et al. [9] proposed BAuth to enhance the authentication security of backscatter communication devices. BAuth is a multistage authentication and attack tracing scheme that applies the backscatter devices' physical spatial attributes, such as RSS, and has high security and performance. Luo et al. [21] proposed ShieldScatter, which uses backscatter to construct a fine-grained physical layer signature against the active attack toward an IoT.

ShieldScatter attaches several tags to construct a sensitive profile, and then it uses a one-class classifier support vector machine (SVM) to detect the attacks according to the received sensitive profile. Huang et al. [22] proposed ScatterID to detect Sybil attackers in ad hoc robotic networks. ScatterID also attaches the backscatter tag to the single-antenna robots and “manipulates” the backscatter signal propagation to create fine-grained lightweight signatures. Li et al. [23] combined ambient backscatter communication and nonorthogonal multiple access (NOMA) to identify the users and prevent illegal access and malicious activities. BAuth focuses on the authentication of homogeneous backscatter communication devices, a significant difference from FedScatter. ShieldScatter and ScatterID focus on lightweight attack detection on resource-constrained wireless devices rather than device authentication. These two studies inspired us to apply backscatter communication to realize a lightweight authentication in FedScatter. Additionally, the backscatter communication-based NOMA symbiotic described in [23] theoretically demonstrates the feasibility of implementing backscatter-based authentication. Instead, we implement a backscatter-based cross-domain authentication in FedScatter.

B. Machine-Learning-Based Authentication

ML-based authentication is a novel approach that supports intelligent and accurate authentication. Authentication schemes for different scenarios widely apply ML, which can effectively learn and track the variations of input data characteristics. Luo et al. [21], Huang et al. [22], and Liu et al. [27] used ML to identify the physical-layer profiles. Fang et al. [17] proposed an intelligent authentication that combines ML and physical-layer attributes, and the authors argue that ML-based intelligent authentication can achieve low cost, high reliability, and continuous and situation-aware device validation under unknown network conditions and unpredictable dynamics. Abdrabou and Gulliver [24] proposed an adaptive, lightweight PLA scheme using ML; the proposed scheme exploits antenna diversity at the receiver to increase the number of features and then trains an SVM-based one-class classifier for outlier and anomaly

detection. Qiu et al. [26] proposed a deep-learning-based authentication scheme, which uses a convolutional-long short-term memory to deal with channel differences without knowing the statistical properties of the channel. Oligeri et al. [18] used deep learning classification algorithms to investigate the effectiveness of low-Earth orbit (LEO) satellites. Their study result shows that a deep-learning-based scheme can successfully authenticate the satellite transducers with a high accuracy. Abdrabou and Gulliver [24] proposed a lightweight PLA to adapt heterogeneous characteristics of wireless mobile networks; they used an SVM with magnitude and frequency characteristic of the received signal as identity features. This scheme requires the receiver to have multiple antennas and is vulnerable to powerful attacks.

FL has unique advantages in privacy-preserving and solving data isolation problems. Yazdinejad et al. [25] proposed an FL-based drone authentication model, which implements the deep neural network (DNN) architecture for local drones and applies homomorphic encryption for securing model parameters aggregation. The experimental results in this work show that FL-based drone authentication gains better performance than other ML-based models. Coelho et al. [29] presented a security management approach for authentication in the Internet of Health Things (IoHT). The presented approach relies on FL for the biometric identification of users while protecting users' privacy. Wazzeah et al. [30] built a novel strategy for continuous authentication based on FL, which protects the user data's anonymity and maintains the data's security. Zhang et al. [31] realized an effective CAN bus anomaly detection system by federated graph neural network to take advantage of crowdsourcing and protect data privacy.

To summarize the above works, applying FL for authentication gains an appealing benefit for privacy, security, and alleviation of model train burden. However, there is still lacks an advanced authentication scheme to support lightweight cross-domain HW-IoTDs authentication. In this article, we design a novel authentication scheme that applies backscatter communication and FL among distributed wireless IoT clusters for lightweight, secure, and accurate authentication of wireless IoT-Ds.

III. FEDSCATTER OVERVIEW

In this section, we first present the system model of FedScatter, including the system entity, threat models, and design goals. Then, we introduce the main technology used in FedScatter.

A. System Model

Our scheme considers three core parties, as shown in Fig. 1, which are distributed HW-IoTDs, edge servers (ESs), and an AS.

HW-IoTD: We consider all IoT-Ds in a specific domain as an IoT cluster. HW-IoTDs, which can be any type with diverse capabilities, are distributed in different IoT clusters. The devices in a cluster attach the same number of backscatter tags for device authentication, and devices in different clusters could attach a diverse number of backscatter

tags according to their hardware resources. When an IoT-D actively communicates with other devices, its backscatter tag absorbs the communication power and passively reflects the signal. Meanwhile, each IoT-D can receive the reflected signal and extract the identity information for authentication. Additionally, FedScatter considers that most IoT-Ds are hardware resource-limit, which makes it hard to collect backscatter communication data and train an authentication model. Therefore, they only provide authentication identity signatures, passively apply the authentication model, and do not participate in the training of the authentication model.

ES: ESs are set up in distributed IoT clusters to generate the global authentication model. In FedScatter, ESs have more computing power and hardware resources than a single IoT-D. The ESs will collect the raw backscatter communication data from the IoT-D and construct several identity vectors, which will be used as a training set for local authentication models. The ESs also communicate with the AS simultaneously and send the parameters of the local model to the AS for aggregation and update the global model with the aggregation results. Finally, the ESs will share the global model with IoT-Ds for authentication between HW-IoTDs.

AS: AS is an independent party. It receives the local model parameters from the ESs and then aggregates them by a customized aggregation algorithm. Finally, AS will return the aggregated model parameters to each ES to allow them to complete the global aggregation. The AS will not collude with any ES or tamper with the local model parameters in the parameter aggregation process.

B. Threat Model

Our system assumes that the ESs and AS are trustworthy parties that can honestly execute the preset programs, such as backscatter signal reception, identity vector generation, and model parameters aggregation. Our main threat comes from wireless IoT-Ds, which could have complex identities in IoT networks. In detail, there are two types of security threats from wireless IoT-Ds. In our work, we face the main identity threat of wireless IoT-Ds. The security of FL itself is not considered.

- 1) *Semi-Honest Attacker:* The attacker would follow predefined communication protocols in the wireless network but expect to gain more network resources. A semi-honest attacker could impersonate an honest device and eavesdrop on sensitive information [32]. Multiple semi-honest attackers also can collude with each other to deceive the honest devices and obtain disproportionate network resources [6]. They could deplete the wireless network resources, eavesdrop on sensitive information, and gain disproportionate benefits.
- 2) *Powerful Attacker:* The attacker would analyze and emulate the communication channel information to break the authentication method [20]. The powerful attackers can also collude to generate a great deal of incorrect authentication information to mislead the honest devices [33], which would also invalidate the authentication method.

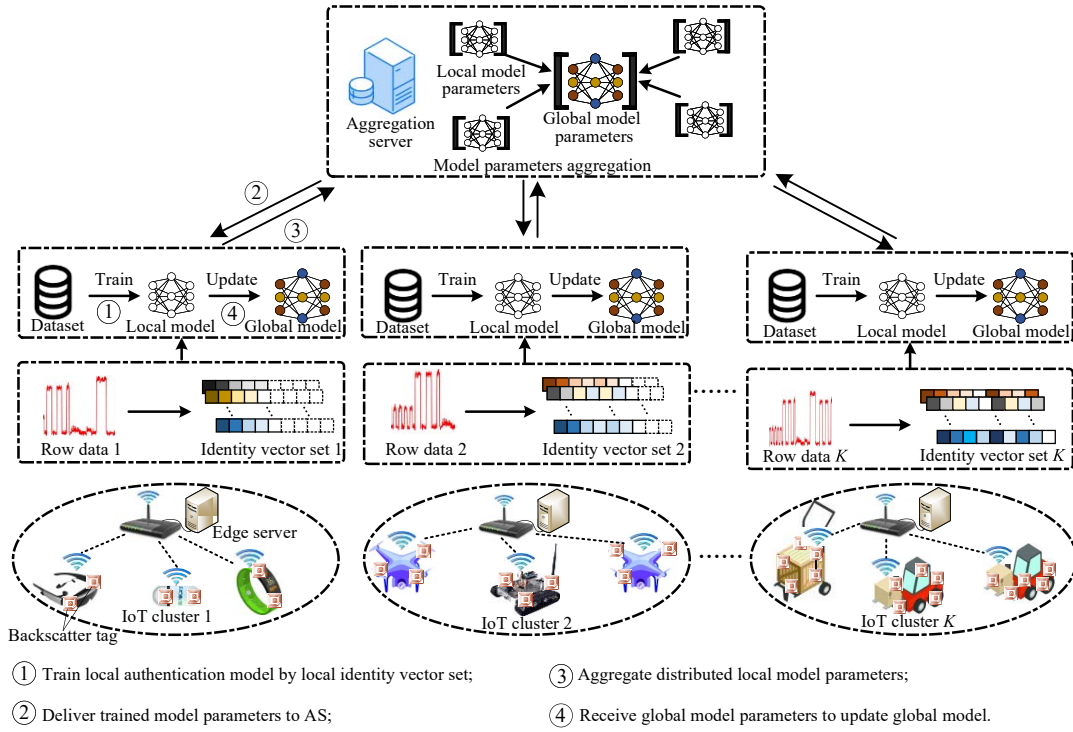


Fig. 1. Illustration of FedScatter authentication scheme.

In FedScatter, the semi-honest and powerful attackers are all in the minority.

C. Design Goal

There are three design goals for the FedScatter.

- 1) *Lightweight*: As illustrated in Section I, most wireless IoTs are hardware resource-limited. Due to the concern of battery and payload, they may not only be unable to perform complex encryption and decryption authentication efficiently but also resist the massive antenna deployment and complex array operation. Therefore, our authentication scheme should not incur additional power and computing consumption to suit resource-constrained HW-IoTDs.
- 2) *Security*: FedScatter should not leak the communication channel information and identity signatures of HW-IoTDs; it also cannot invalidate even in the presence of a powerful attacker. Moreover, FedScatter is required to implement continued protection during the HW-IoTDs communication process, as described in [17]
- 3) *Accuracy*: In the absence of an attack, the authentication accuracy of FedScatter cannot be sacrificed. FedScatter should also accurately discriminate the legitimacy of wireless IoT identities to prevent attacks from various attackers.

D. Related Technology

1) *Backscatter Communication*: Backscatter communication is a novel battery-free and ultralow-power communication paradigm, as shown in Fig. 2. The backscatter tag can passively scatter spatial active radio-frequency signals to perform

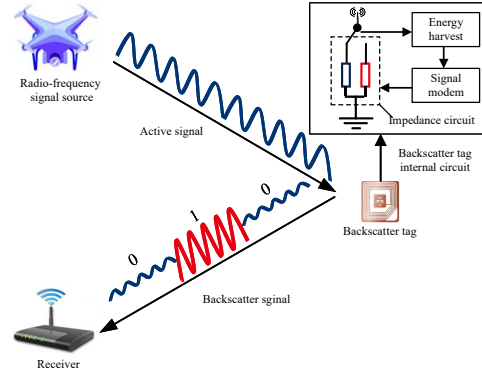


Fig. 2. Backscatter communication schematic.

data transmission without the need for power-hungry and bulky radio components [34]. The power consumption of a backscatter tag can be as low as μW level, and the hardware cost is as cheap as a few U.S. cents [35].

Backscatter-based signal propagation is generated by manipulating the signal propagation of multiple backscatter tags. According to [36], a receiver can receive an active communicate signal and a passive backscatter signal simultaneously, which can be expressed as

$$y(n) = y_t(n) + ky_b(n)y_t(n) + \eta(n) \quad (1)$$

where $y(n)$ is the final signal sequence received by the receiver, $y_t(n)$ is the active communicate signal sequence from the random forest (RF) signal source, k is the reflection attenuation coefficient which is related with impedance circuit of backscatter tag, $y_b(n)$ is the reflect control sequence which has two states $\{0, 1\}$, and $\eta(n)$ is the propagation noise.

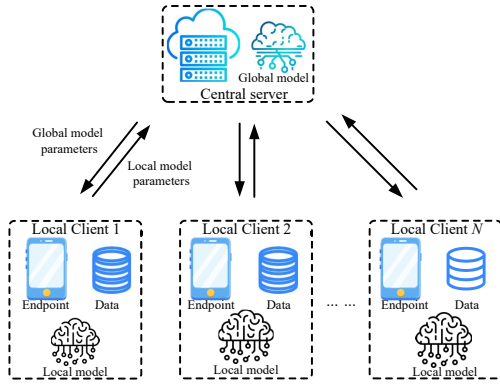


Fig. 3. Architecture of typical FL.

When multiple backscatter tags are present around the RF signal source, the reflection attenuation coefficient vector and reflect control sequence vector are expressed as $\mathbf{k} = (k_1, k_2, \dots, k_i)$ and $\mathbf{y}_b(n) = (y_{1b}(n), y_{2b}(n), \dots, y_{ib}(n))$, respectively, where i is the number of backscatter tags.

The backscatter signal propagation is sensitive to the active signal strength, distance, and relative distance [36]. Furthermore, the average energy of backscatter signal propagation from the same set of backscatter tags would have a substantial similarity, which is similar to AoA signatures [21]. In other words, when multiple backscatter tags are attached to an IoT, the multipath signal propagation is highly correlated with the spatial attributes of the IoTs and contains authenticatable identity information.

In FedScatter, we use the multiple backscatter tag to build identity signatures for IoT authentication, which realize the light authentication among wireless IoTs and against the attack from semi-honest and powerful attackers.

2) *Federated Learning*: FL is first introduced by Google [37], which aims to apply large amounts of valuable data that does not violate privacy. It allows all participants to train robust models despite insufficient training data and can alleviate the training burden compared with central deep learning [26]. The mainstream FL scheme includes horizontal FL, vertical FL, and federated transfer learning [38]. A typical FL has three main phases: 1) local model training; 2) model parameters aggregation; and 3) global model update, as shown in Fig. 3.

All participants in FL collaboratively train a global model without sharing their local data. Each participant uses their local data to train a local with an arbitrary neural network in the local training phase. The participant will transmit their local model parameters to the central server in the model aggregation phase. The central server receives most of the model parameters from distributed participants and then aggregates them via a predefined aggregation algorithm. Finally, the central server will transmit the aggregated global model parameters to each participant in the global model update phase. The participants update their local model by the global model parameters and generate a global model.

In FedScatter, we use the FL to protect the identity signature information of different wireless IoT clusters against the attack

from powerful attackers, which can make the authentication fail. Furthermore, the FL can also alleviate the data storage and computing load of the ESs in FedScatter.

IV. FEDSCATTER DESIGN

We summarize the FedScatter design in Fig. 4 and describe the details of FedScatter design, including authentication flow, identity signature construction, and universal authentication model generation.

A. Identity Signature Construction

In FedScatter, the distributed ESs among different IoT clusters will selectively collect the backscatter signal to construct the identity signature to deliver authenticatable identity information for HW-IoTDs. Meanwhile, the ESs also use the signatures to generate data sets for universal authentication models.

1) *Basic Identity Signature*: According to [36], when a backscatter tag is fixed, the backscatter signal strength $A_{\text{backscatter}}$ is highly related to the 3-D Euclidean distance between the tag and the transmitter, as expressed in

$$A_{\text{backscatter}} = G \times \frac{k_1}{4\pi d_{tx}^2} \times \frac{k_2}{16\pi^2 d_{rx}^2} \quad (2)$$

where G , k_1 , and k_2 are the gains of environments, transmitting antenna, and receiving antenna, respectively. d_{tx} and d_{rx} are the 3-D Euclidean distance of the backscatter tag. In FedScatter, for the honest wireless IoTs, G , k_1 , and k_2 will be the constant values. Therefore, the fixed tags mainly reflect the host IoT signal without significantly interfering with other IoTs. We assume that n backscatter tags are fixed to an IoT. For an ES, it received the backscatter signal sequence of the IoT can be expressed as

$$\{s_1, \dots, s_i, \dots, s_n\} \quad (3)$$

where s_i denotes the signal time-sequence vector generated by the i th backscatter tag. For alleviate the other unwanted reflections in s_i , we used the method proposed in [22], which can be expressed as

$$c_i = \left| \frac{1}{N_r} \sum_{t=1}^{N_r} s_i(t|\text{ref}) - \frac{1}{N_n} \sum_{t=1}^{N_n} s_i(t|\text{nonRef}) \right| \quad (4)$$

where c_i is the reflect signal average value of the i th backscatter tag, N_r and $s_i(t|\text{ref})$ are the reflected sample size and t index vector value, and N_n and $s_i(t|\text{nonRef})$ are the nonreflected sample size and t index vector value. Then, for n backscatter tags are fixed on a wireless IoT, we can construct a basic identity signature $\mathbf{v}$ which provided a high related based on (3) and (4)

$$\mathbf{v} = (c_1, \dots, c_i, \dots, c_n). \quad (5)$$

Meanwhile, we denote $\mathbf{v}^j, j \in \{1, 2, \dots, m\}$, as a basic identity signature of the j th wireless IoTs in an IoT cluster with

Identity Signature Construction:

- 1) *Basic identity signature*: The ES in an IoT cluster receives multiple backscatter communication signals from an IoT, processing and extracting these features. Then, constructing an essential identity signature for HW-IoTD.
- 2) *Supplementary identity signature*: To authenticate the mobile wireless IoT, and against the attack accuracy from the semi-honest IoT, the ES will supplement the essential identity signature with a constant backscatter signal characteristic.

Universal Authentication Model Generation:

- 1) *Local model training*: The ES in an IoT cluster trains a local model using wireless IoTs' local identity signatures dataset. Model parameters include neural network structure, loss function, number of local iterations, and other hyperparameters.
- 2) *Global model parameters aggregation and update*: When the distributed local model has completed its pre-specified iterations, the AS will receive the model parameters, including the identity signatures information of the HW-IoTDs, and aggregate these model parameters by a customized aggregation algorithm. The AS returns the aggregated global model parameters to the distributed ESs for the global model update. Then, generate a universal authentication model for HW-IoTDs.

Device Authentication Flow:

- 1) *Pre-authentication*: The ESs build the identity signatures and train a universal authentication model through the methods described in Sec. IV-A and Sec. IV-B. Then, the ESs send the trained models to wireless IoTs in their clusters. The devices deploy the models for authentication.
- 2) *Authentication*: The device authentication flow includes access authentication and IoT peer-to-peer authentication. The former is that the ES authenticates the identity of a wireless IoT, and the latter is that a HW-IoTD authenticates another HW-IoTD. Additionally, FedScatter optimized authentication among the HW-IoTDs, which will be discussed in detail in IV-C.

Fig. 4. FedScatter overview for HW-IoTDs' authentication.

m wireless IoTs. An ES in the cluster could build a basic identity signature set $\mathbf{D}$ as

$$\mathbf{D} = \begin{bmatrix} \mathbf{v}^1 \\ \vdots \\ \mathbf{v}^j \\ \vdots \\ \mathbf{v}^m \end{bmatrix} = \begin{bmatrix} c_1^1 & \cdots & c_i^1 & \cdots & c_n^1 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ c_1^j & \cdots & c_i^j & \cdots & c_n^j \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ c_1^m & \cdots & c_i^m & \cdots & c_n^m \end{bmatrix}. \quad (6)$$

2) *Supplementary Identity Signature*: To this end, the wireless IoTs in FedScatter can use the identity set $\mathbf{D}$ for the static device authentication. However, the wireless IoT may also be moving, which means that the spatial attributes of the backscatter signal will change continuously. Then, the authentication accuracy of the identity set $\mathbf{D}$ may decrease. Meanwhile, as illustrated in Section III-B, a semi-honest wireless IoT could also make a new fake identity by changing its antenna transmit power or colluding by sharing the respective identity signatures.

In order to overcome the above difficulties, we use a cosine similarity of two round backscatter signals as an additional feature. The value of the essential identity signature will be changed because of the change of spatial state between the backscatter tags and the receiver, whether the moving state, transmission power change, or colluding of semi-honest IoTs. Therefore, we extend the base identity signature vector $\mathbf{v}^j$ to ensure that the j th IoTs can achieve accurate authentication even in arbitrary environments. We consider that the backscatter tags are fixed on the IoTs, which means that

identity signatures generated by several backscatter tags have unique and constant spatial attributes related to the IoTs. The cosine similarity, as a measure of the difference between two identity signature vectors from IoTs, can reflect the changes in spatial attributes. Formally, the cosine similarity $Siml$ is defined as

$$Siml(\mathbf{v}^j, \mathbf{v}^{j'}) = \cos\theta = \frac{\langle \mathbf{v}^j, \mathbf{v}^{j'} \rangle}{\|\mathbf{v}^j\| \cdot \|\mathbf{v}^{j'}\|} \quad (7)$$

where $\mathbf{v}^j$ is the first constructed identity signature vector of the j th wireless IoT, $\mathbf{v}^{j'}$ is the second, and θ is the angular difference between the identity signature vector $\mathbf{v}^j$ and $\mathbf{v}^{j'}$. $\langle \cdot, \cdot \rangle$ is the inner product operation. $\|\cdot\|$ is the L-2 norm operation for $\cdot$. Using cosine similarity makes FedScatter effective in IoTs moving states and mitigates attacks from semi-honest devices. The expanded identity signature set $\mathbf{D}'$ for an IoT cluster with m IoTs is defined as

$$\mathbf{D}' = \begin{bmatrix} \mathbf{v}^1 & \mathbf{v}^{1'} & Siml^1 \\ \vdots & \vdots & \vdots \\ \mathbf{v}^j & \mathbf{v}^{j'} & Siml^j \\ \vdots & \vdots & \vdots \\ \mathbf{v}^m & \mathbf{v}^{m'} & Siml^m \end{bmatrix} \quad (8)$$

the ES will build the $\mathbf{D}'$ in several times as a part of the local data set for a universal authentication model.

B. Universal Authentication Model Generation

Section IV-A illustrates an IoT cluster's identity signature construction scheme. Without loss of generality, the HW-IoTDs distributed in different IoT clusters and can also use the scheme to construct backscatter-based identity signatures. In FedScatter, we expect a universal model $\mathcal{M}$, which can authenticate the legitimacy of HW-IoTDs by the given identity signatures and secure the devices from powerful attackers. However, for HW-IoTDs, the devices may be equipped with different number of backscatter tags, which could hinder the construction of an identity signatures data set. The FL support heterogeneous data since the ES in different IoT cluster can use local data set. Therefore, we use FL to generate $\mathcal{M}$ and customize a model parameters aggregation strategy. We also facilitate knowledge sharing of authentication information and enhance device authentication effectiveness between the HW-IoTDs through FL.

1) *Local Model Training*: We first train a local model for each IoT cluster $k, k \in \{1, 2, \dots, K\}$. As illustrated in Section IV-A, the ES in an IoT cluster collects the identity signatures of wireless IoT clusters as part of the data set for the local model training. Focusing on the backscatter-based identity signatures set $\mathbf{D}_k$ in any IoT cluster k . We set the labeled data set $\mathcal{C}_k = \{(\mathbf{D}_k^m, \mathbf{y}_k^m)\}_N$, where N is the samples size and $\mathbf{y}_k^m \in \{0, 1\}_N^m$ is the related authentication label that 1 denotes the valid identity signatures, and 0 denotes the others.

Then, IoT cluster k trains its local model M_i with its local authentication data set $\mathcal{C}_k$ and then iteratively optimizes the local model ω_k^t during each epoch t that is expressed as

$$\omega_i^{t+1} = \omega_i^t + \eta \nabla \omega_i^t \quad (9)$$

where η is the learning rate, and $\nabla \omega_i^t = \nabla \omega_i^t(x, \xi_i)$ is a stochastic gradient of the loss function. In FedScatter, we use the cross-entropy loss function, which is expressed as

$$\text{loss}_{\text{crossEntropy}} = - \sum_{i=1}^n y \log(\hat{y}) \quad (10)$$

where y is the true label, and $\hat{y}$ is the predicted label.

After most wireless IoT clusters train the local model iteratively several times, they will send the local model weight parameters to the AS for global model parameters aggregation.

2) *Global Model Parameters Aggregation and Update*: In this phase, the AS will aggregate the local model parameters from the ESs for a global optimization goal. The goal is formulated as follows:

$$\min_{x \in \mathbb{R}^d} \mathcal{M}(x) = \min_{x \in \mathbb{R}^d} \sum_{s=1}^n \frac{1}{n} \mathcal{M}(x) \quad (11)$$

where $\mathcal{M}(x)$ is the global loss function in FedScatter, $\mathcal{M}(x) := \mathbb{E}_{\xi_s}[\mathcal{M}(x, \xi_s)]$, and ξ_s is the loss function and the test sample of each IoT cluster, respectively. $\mathbb{R}^d$ is the global samples data set among all IoT cluster.

Generally, the model parameters aggregation process relies on an aggregation algorithm for generating global model parameters, such as FedAvg [37], the first aggregation algorithm that assigns local model weights by the size of the local model data set. However, the FedAvg used in FedScatter fails

Algorithm 1: Universal Authentication Model Construction in FedScatter

Input: Distributed local data set $\mathcal{C}_k$, the tags number in each IoT cluster b_k , for $k \in \{1, 2, \dots, K\}$, iteration times T , learning rate η .

Output: Global model parameters ω^t

```

1 begin
2   t := 0;
3   Local Training:
4   for each ES in distributed IoT cluster  $k = 1$  to  $K$  in parallel do
5     while  $t < T$  do
6       if  $t = 0$  then
7         initializes model weight  $\omega_k^t$ 
8         start training based on local signatures dataset  $\mathcal{C}_k$ 
9       else
10        computation:  $\nabla \omega_k^t$ 
11         $\omega_i^{t+1} = \omega_i^t + \eta \nabla \omega_i^t$ 
12      end
13       $t = t + 1$ 
14    end
15    transmit the  $\{\omega_k^t, b_k\}$  to the AS.
16  end
17  Parameters Aggregation:
18  The AS do
19     $B = \sum_{k=1}^K b_k$ 
20     $\omega^t = \sum_{k=1}^K \frac{b_k}{B} \omega_k^t$ 
21    transmits  $\omega^t$  to each ES in distributed IoT clusters.
22  return  $\omega^t$ 
23 end

```

to reflect the authentication characteristics of the heterogeneous IoT clusters. The power attack could also create large amounts of data to disrupt the construction of authentication models. We observe that the number of backscatter tags affixed on the device influenced the effect of the attack detection [22], while the attacker cannot forge the actual tags. Then, we heuristically design a customized aggregation algorithm that configures all weights of the local model parameters for the global model by the number of backscatter tags rather than by data set size. Algorithm 1 illustrates our algorithm and the overall process of FedScatter universal authentication model generation.

C. Authentication Flow

We describe specific schemes in detail about the authentication identity signatures and universal model of FedScatter above. Now, we present the specific authentication flow of FedScatter, as shown in Fig. 5. The authentication flow of wireless IoT clusters is divided into a preauthentication phase and an authentication phase.

1) *Preauthentication*: In FedScatter, all HW-IoTDs that comply with the authentication protocol can be seen as semi-honest devices, which expect access to disproportionate resources via malicious attacks after gaining the trust of other



Fig. 5. FedScatter authentication flow.

devices. In other words, the semi-honest devices will honestly provide their identity signature to the ES via their backscatter tags during the preauthentication phase and launch attacks against other devices or communication networks during the authentication phase.

The ES in a cluster first starts communicating with wireless IoT devices and requests the devices to respond. A wireless IoT device responds to the request of ES with a signal, which combines the active communication signal and the passive signal generated by backscatter tags. Then, the ES receives the signals, builds, and stores the device's identity signature via the method described in Section IV-A. After the ES stores the identity signatures of all the wireless IoT devices in its cluster, it will train a local authentication model via the method described in Section IV-B1.

While the ESs complete the local authentication model training, they send the local model parameters to the AS. The AS aggregates the parameters by the aggregate algorithm described in Algorithm 1 for a global model. The AS sends the global model parameters to the ESs, participating in the parameters aggregation process. The ESs receive the global model parameters and update the model for a universal authentication model. Finally, the ESs distribute the universal authentication model to wireless IoT devices in the same cluster, and the wireless IoT devices deploy the model for IoT peer-to-peer authentication.

2) *Authentication*: As shown in Fig. 5, the authentication phase in FedScatter supports two authentication flows for IoT peer-to-peer authentication and access authentication and consists of basic and continuous authentication. The ESs store the backscatter-based identity signature information in the preauthentication phase, which means a wireless IoT device with backscatter tags is considered as *registered* device with the

ES, and IoT devices cannot obtain the *registered* information from the ES for privacy concern. Therefore, FedScatter designs two schemes in basic authentication subflow authentication flow, the IoT authentication and the ES authentication, for different authentication flows.

The IoT authentication is a wireless IoT device that authenticates another device that can support HW-IoTDs from different domains and defend against attacks from semi-honest devices. In the basic authentication subflow, the wireless IoT device i receives the first round communication signal sequence from another device j . It only needs to extract the backscatter-based identity signature v_j^1 of the device via (3) and (4) and store the signature for the next authentication.

The ES authenticates the wireless IoT devices in the same IoT cluster for IoT network security. In the basic authentication subflow, the ES receives a communication signal sequence of an IoT device, and it extracts the first backscatter-based identity signature v^1 of the IoT device via (4) and (5). Then, the ES will compute the cosine similarity with the signature vector in stored identity set $\mathbf{D}$ via (7) to ensure whether the device is a *registered* device during the precertification phase.

In the continuous authentication subflow, the device i receives the t th round communication signal sequence and extracts the signature. For identity signature v_j^{t-1} and v_j^t of $t-1$ th and t th round communication. Then, the device i computes cosine similarity Siml of v_j^{t-1} and v_j^t via (7) and generates an authentication vector $\mathbf{v}' = (v_j^{t-1}, v_j^t, \text{Siml})$. Finally, the device i inputs $\mathbf{v}'$ to the universal authentication model to check the legitimacy of the identity of the device j . The authentication result is expressed as follows:

$$\mathcal{M}(\mathbf{v}') = \begin{cases} 1, & \text{legitimate IoT device} \\ 0, & \text{illegitimate IoT device.} \end{cases} \quad (12)$$

Using FedScatter, the HW-IoTDs can construct an identity signature by attaching lightweight, battery-free, ultralow power backscatter tags. Then, an ES or a wireless IoT device can authenticate arbitrary IoT devices in FedScatter, and vice versa. By extracting the signature information from the communication signals based on backscatter tags, the ESs or IoT devices input the signature information to the universal authentication model and gain the authentication result.

V. SECURITY ANALYSIS

In this section, we analyze the security of FedScatter under the semi-honest attack and the power attack.

A. Attack From Semi-Honest Attackers

As described in Section III, the semi-honest attackers will follow the network's communication protocols but expect to gain access to more resources. These attackers launch the spoofing attack, the Sybil attack, the replay attack, and the man-in-the-middle attack in the IoT network to gain access to additional communications resources. Therefore, we discuss how FedScatter implements defense against these four attacks to demonstrate its security under the semi-honest attacks.

1) *Spoofing Attack Defense*: In the considered spoofing attack, it is assumed that the attacker possesses knowledge of the authentication scheme and the identities of other legitimate devices. The attacker attempts to impersonate a legitimate device and spoof another honest device. Based on (2) and (4), we extract backscatter signal features of the wireless IoTDS, which are highly correlated with the spatial distribution of backscatter tags and Euclidean distance between the transmitter, backscatter tags, and receiver. Then, we use the features to construct the identity signature for wireless IoTDS authentication. Due to the discrepancy in the distribution of backscatter tags and position between the attacker and other devices, the attacker cannot mimic the identity signature of other honest devices. Besides, we utilize a continued authentication scheme to detect the identity legitimacy of a mobile wireless IoTDS. Thus, FedScatter can defend against spoofing attacks when an attacker impersonates a legitimate wireless IoTDS identity.

2) *Sybil Attack Defense*: Sybil attack is a multi-identity-based attack method in a single device with multiple identities in a network. In a PHY-based authentication scenario, a Sybil attacker also changes its antenna transmitting power for forging identities. In FedScatter, a wireless IoTDS's passive signal propagation is identical due to its fixed spatial distribution of backscatter tags and distinctive position. Each wireless IoTDS's backscatter signal traces correlate throughout the communication epoch. Based on (5), (7), and (8), we utilize the correlation to construct a unique identity signature for each HW-IoTD. Even if a Sybil attacker forged a fake identity to communicate with ES or other honest HW-IoTD, the receiver can detect its actual identity signatures by the passive backscatter signal. Besides, we use the passive signal similarity measure method over two rounds of communication. In this way, the ESs or honest HW-IoTDs can detect whether the attacker changes its antenna transmitting power or conspires with another device to share an identity. Thus, the devices in FedScatter can defend against multiple Sybil attacks.

3) *Replay Attack Defense*: A signal replay attacker endeavors to capture target HW-IoTD signals and replays them for device authentication. In FedScatter, the target HW-IoTD's signals include active communication signals and passive backscatter signals. The active communication signal emanated by the device's antennas is used to transmit a message, and the passive backscatter signals produced by the backscatter tags are used to construct the device's identity signatures. Thus, in the event of a relay attack, an attacker can only record the active communication signals and replay the recorded signal to the ES or other HW-IoTDs; the receiver can detect this attack attempt because the passive backscatter signal of the attacker is different from the previously HW-IoTD. Thus, the receiver can terminate the communication epoch with the attacker to defend against the replay attack.

4) *Man-in-the-Middle Attack Defense*: A man-in-the-middle attacker [39] can intercept and potentially alter the correspondence between the transmitter and receiver, leading both parties to trust that they are communicating with each other straightforwardly. It is assumed that, in FedScatter, a man-in-the-middle attacker manipulates the communication

link between the transmitter and receiver of HW-IoTs. However, the receiver will detect the alterations of identity signature from the passive backscatter signal of the transmitter, and vice versa. Consequently, a man-in-the-middle attacker cannot succeed in FedScatter.

B. Attack From Powerful Attackers

As described in Section III, a powerful attacker in an IoT cluster will hold the identity signatures set $\mathbf{D}'$ and labeled data set $\mathcal{C}_k$ to know the authentication rules in FedScatter. Then, the powerful attacker can use its ample hardware resources or collude with other devices to generate amounts of incorrect identity signature samples to ES for an adverse local model to undermine the authentication scheme. FedScatter is resilient to powerful attackers due to FL with Algorithm 1. FedScatter applies a *backscatter tags determine weight* parameters aggregation algorithm, preventing IoT clusters from controlling the global common authentication models. As demonstrated in Section V-A, an IoTDS's identity signature remains unassailable due to the passive construction of its identity signatures, leading to powerful attackers not emulating other legitimate IoTDSs. In this way, the powerful attacker cannot generate spurious identity signatures to mislead the ES. Even if several powerful attackers in an IoT cluster collude to produce an adverse local model, the limited weight based on Algorithm 1 can alleviate the effect of the adverse model.

VI. IMPLEMENTATION AND EVALUATION

In this section, we evaluate the performance of FedScatter. First, we evaluate the impact of the number of backscatters on identity signatures for wireless IoTDS authentication. Then, we evaluate the performance of our proposed universal authentication model and aggregation algorithm in FedScatter. Finally, we evaluate the performance of FedScatter in different attack scenarios.

A. Implementation

We implement FedScatter using several backscatter tags, USRP-2954R, and a server, as shown in Fig. 6. We deploy several backscatter tags surrounding the USRP nodes as heterogeneous IoTDSs with different hardware resources. In particular, the backscatter tags use AGLN250 FPGA for the processing of data to be launched, and it is equipped with an ADG902 radio switch to adjust its own reflection coefficient. We set the USRP parameters as 2.4 GHz of work frequency and 20 MHz of bandwidth. Meanwhile, to prevent interference between the active communication signal and the passive backscatter signal, we set the 40-MHz frequency shift. We also use a USRP-2954R connected to a computer as the ES. The USRP is set as a receiver, and we also set the USRP parameters as 2.4 GHz of work frequency and 20 MHz of bandwidth. The computer will process the backscatter signal as described in Section IV-A and train the local authentication model as described in Section IV-B. The AS is a computer software that includes the parameters aggregation algorithm described in Algorithm 1.

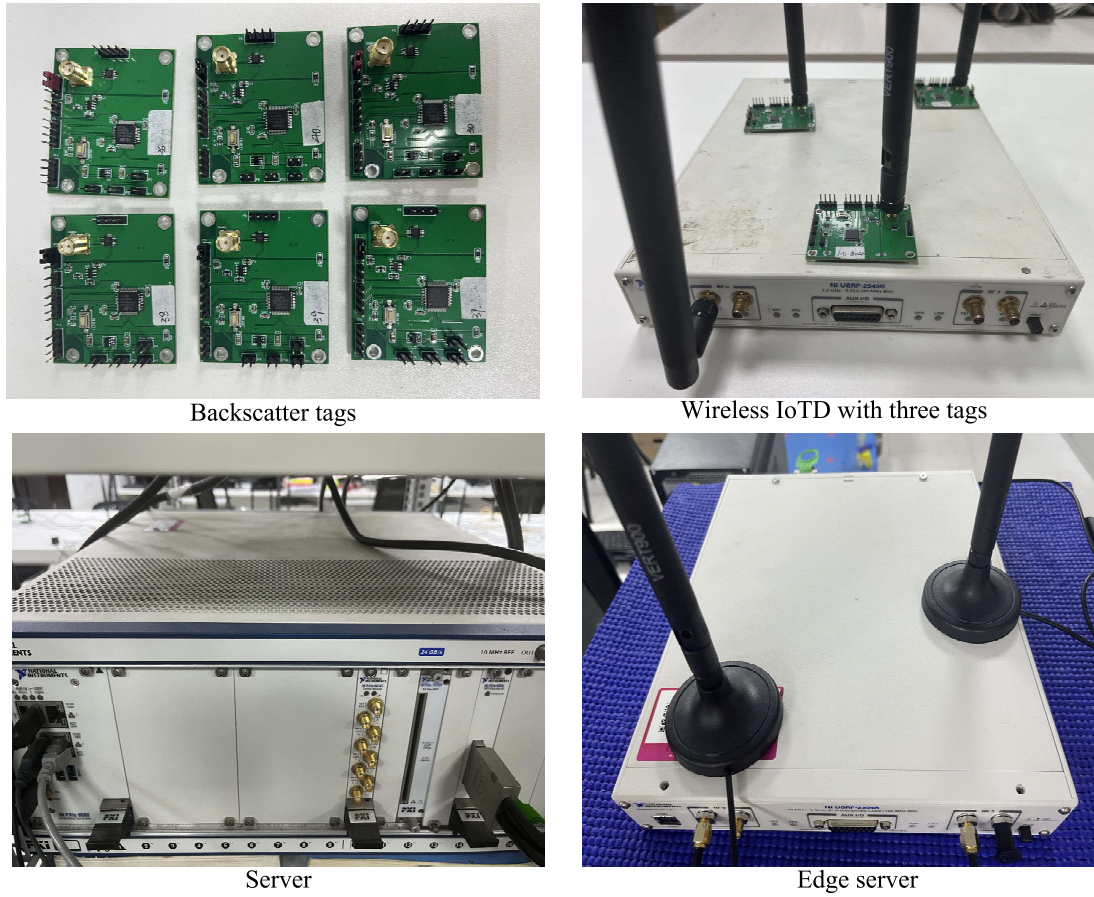


Fig. 6. Experimental platform. We implement FedScatter using several backscatter tags, USRP devices, and a server.

We use the following metrics to evaluate the performance of the FedScatter authentication scheme.

- 1) *Authentication Accuracy*: It is the ratio of the number of wireless IoT devices that are correctly authenticated to the number of all wireless IoT devices.
- 2) *TPR*: It is the ratio of legitimate wireless IoT devices that FedScatter successfully authenticates to the total number of legitimate wireless IoT devices.
- 3) *FPR*: It is the ratio of the number of illegitimate IoT devices authenticated as legitimate wireless IoT devices to the total number of illegitimate wireless IoT devices.

Additionally, we also use receiver operating characteristic (ROC) curve and area under the ROC curve (AUROC) to evaluate the comprehensive performance of classifiers in FedScatter.

B. Experimental Results

1) *Performance of Backscatter-Based Authentication*: The key to authenticating a wireless IoT device relies on deploying backscatter tags. As described in (5) and (7), the identity signatures consist of multibackscatter signal propagation. According to Sections IV-B and IV-C2, the dimension of the identity signatures vector in a wireless IoT cluster depends on the backscatter tag fixed on the IoT devices. Therefore, we first evaluate the authentication performance of different numbers

of backscatter tags on the IoT devices of the local model in an IoT cluster. We use the DNN for the local authentication model described in Section IV-B. Meanwhile, we choose the classic ML method to compare with our scheme, i.e., RF, decision tree (DT), logistic regression (LR), SVM, and naive Bayes (NB). We summarize the evaluation results in Table II.

The results show that our backscatter-based authentication scheme is valid in a wireless IoT cluster. Table II shows that our DNN model obtains higher TPR and lower FPR overall. The DNN can achieve the highest accuracy of 98.35%, the highest TPR of 97.01%, and the FPR of 0.73%, and the NB achieves the lowest FPR of 0.54% when five backscatter tags contribute to the identity signatures. The worst performance of authentication occurs in identity signatures based on two backscatter tags, which achieve the lowest accuracy of 83.11%, the highest FPR of 17.70%, and the lowest TPR of 77.91% from DT and SVM, respectively. Meanwhile, we observe that the FPR from different machine-learning schemes, which means the ratio of illegitimate wireless IoT devices passing the authentication, declines significantly as the number of backscatter tags rises. In our experiment, the overall authentication performance also improves with the increase in backscatter tags because more tags can provide more fine-grained identity signature features for authentication.

Constructing identity signatures via plenty of backscatter tags and training an authentication model can achieve desirable

TABLE II
AUTHENTICATION PERFORMANCE OF BACKSCATTER-BASED IDENTITY SIGNATURE UNDER DIFFERENT ML METHODS. (A) PERFORMANCE OF TWO BACKSCATTER TAGS-BASED IDENTITY SIGNATURE. (B) PERFORMANCE OF THREE BACKSCATTER TAGS-BASED IDENTITY SIGNATURE. (C) PERFORMANCE OF FOUR BACKSCATTER TAGS-BASED IDENTITY SIGNATURE. (D) PERFORMANCE OF FIVE BACKSCATTER TAGS-BASED IDENTITY SIGNATURE

(a)

	DNN	RF	DT	LR	SVM	NB
Accuracy	0.8668	0.8676	0.8311	0.8713	0.8348	0.8638
FPR	0.1535	0.1449	0.1770	0.1707	0.0956	0.1359
TPR	0.8083	0.8271	0.7803	0.9028	0.7791	0.8621

(b)

	DNN	RF	DT	LR	SVM	NB
Accuracy	0.8946	0.8714	0.8590	0.8877	0.8811	0.8621
FPR	0.0482	0.0509	0.0760	0.0492	0.0852	0.0461
TPR	0.9341	0.8884	0.8741	0.9158	0.8545	0.8727

(c)

	DNN	RF	DT	LR	SVM	NB
Accuracy	0.9714	0.9763	0.9462	0.957	0.9674	0.9714
FPR	0.0107	0.0111	0.0392	0.0391	0.0251	0.0107
TPR	0.9186	0.9173	0.8782	0.9387	0.9320	0.8602

(d)

	DNN	RF	DT	LR	SVM	NB
Accuracy	0.9835	0.9769	0.9660	0.9771	0.9843	0.9791
FPR	0.0073	0.0059	0.0239	0.0204	0.0123	0.0054
TPR	0.9701	0.9501	0.9147	0.9641	0.9668	0.9000

authentication in a wireless IoT cluster. However, deploying more backscatter tags is difficult for IoTDs with limited hardware and computing resources. Meanwhile, the powerful attacker in a wireless IoT cluster will have more influence to destroy the local authentication model. Next, we will illustrate the improvements in performance and security of authentication using FL as the generation scheme of a universal authentication model than central learning in Sections VI-B2 and VI-B3

2) *Performance of FedScatter*: In the above section, we illustrate the performance of backscatter-based authentication. we set four heterogeneous wireless IoT clusters, and the IoT in each cluster has a different number of backscatter tags for identity signatures. Each wireless IoT cluster uses the same identity signature construction method described in Section IV-A and a consistent DNN model.

Analysis Communication Cost and Computing Costs: In FedScatter, the identity signature of wireless IoTDs is constructed by the passive signal and does not consume an additional communication resource. The main communication cost comes from the aggregation of model parameters in FL. Moreover, the construction of an identity signature will consume additional computing costs. We compare the communication and computing costs with the existing scheme [22]. The result is shown in Fig. 7, where communication cost includes model sharing and parameter aggregation, and the computing costs include identity signature construction and device authentication. With the number of IoTDs growing, the communication cost and computing costs of FedScatter are stable and are lower than [22], which means that FedScatter can support the authentication of a large number of wireless

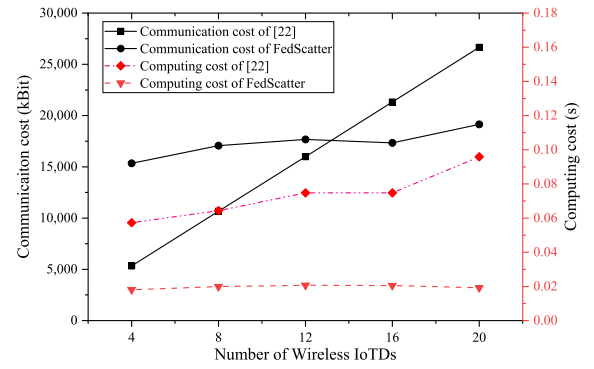


Fig. 7. Communication cost and computing cost of FedScatter compared with [22].

IoTds. Although the FL process brings additional communication costs, the wireless IoTds are not involved in the FL process in FedScatter.

Effectiveness of FedScatter: Next, we demonstrate the effectiveness of our FedScatter for HW-IoTD authentication. We use a FedAvg-based FL method as a reference scheme for the universal authentication model, and FedScatter uses the method described in Section IV-B. We record the global model's accuracy and loss in thirty epochs. We also record the accuracy and loss of the local train model for different IoT clusters synchronically. We evaluate our scheme in a scenario without the attackers.

Fig. 8 shows the authentication performance of the FedScatter model, the model of reference scheme, and four local train models. As the figure described, the model in the cluster with more backscatter tags maintains a high level of accuracy and a low level of loss. FedScatter and FedAvg-based FL schemes also achieve high performance with an accuracy of more than 97% and a loss of less than 0.1, close to the accuracy of wireless IoT clusters with four tags and five tags. Since the universal authentication model is applied to arbitrary IoT clusters, FedScatter can improve the authentication accuracy of IoT clusters where the identity signatures of the wireless IoTds are constructed with few backscatter tags. In more detail, FedScatter has a higher accuracy than the baseline schemes. The primary reason is that FedScatter constructs a backscatter-based identity signature for wireless IoTds. The identity signature is highly correlated with the spatial distribution of backscatter on wireless IoTds. As evaluated in Section VI-B1, more backscatter tags can provide more fine-grained identity signature information. Algorithm 1 uses the parameters aggregation scheme based on the number of backscatter tags. Then, the identity signature information among heterogeneous IoT clusters can be shared more efficiently than the baseline scheme in FedScatter.

3) *Performance of FedScatter Under the Semi-Honest Attacks and Powerful Attacks*: We have discussed the security of FedScatter under the semi-honest attackers and powerful attackers in Section V. Now, we verify that our system resists various attacks through the numerical results. We compare the performance of our scheme with existing work [22], which is applied in ad hoc robot networks to defend against malicious

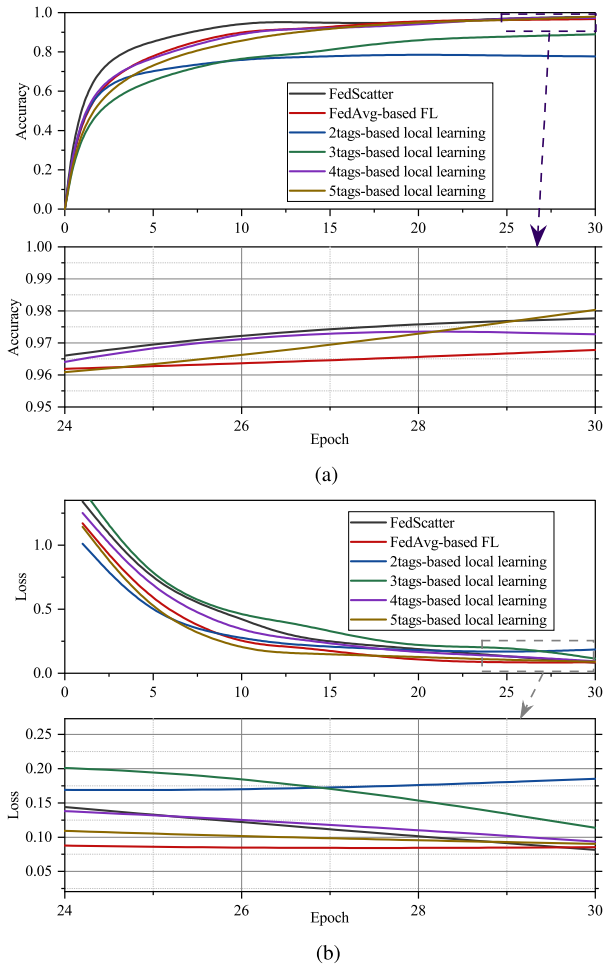


Fig. 8. Evaluation result of FedScatter and five reference schemes in normal scenario. (a) Accuracy of FedScatter and five baseline schemes. (b) Loss of FedScatter and five baseline schemes.

attacks by using backscatter. Additionally, we design two baseline schemes to illustrate the validity of the schemes proposed in our FedScatter.

For the semi-honest attacker experiments, we set up two extra baseline schemes that use the backscatter signals as the identity signatures and generate the universal authentication model by the FedAvg parameters aggregation algorithm. Specifically, baseline scheme 2 applies the supplemental cosine similarity feature, while baseline scheme 3 is without the cosine similarity feature. FedScatter and the baseline schemes will train a universal authentication model according to the process in Section IV, and the test samples will include the fake identity signatures and the actual identity signatures from the moving wireless IoT devices. We also set up two extra baseline schemes for the powerful attackers. Baseline scheme 2 uses the FL to construct the authentication model, and baseline scheme 3 uses central learning. Moreover, the train samples in an IoT cluster will be filled with incorrect labels and several times the size of the other samples. The test samples in powerful attacker's experiments will be the actual identity signatures of wireless IoT devices.

Fig. 9 illustrates the authentication performance of FedScatter and the baseline schemes under the semi-honest

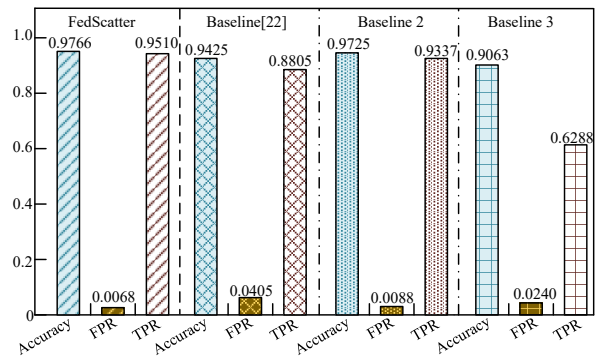


Fig. 9. Performance of FedScatter and three baseline schemes under semi-honest attacks scenario.

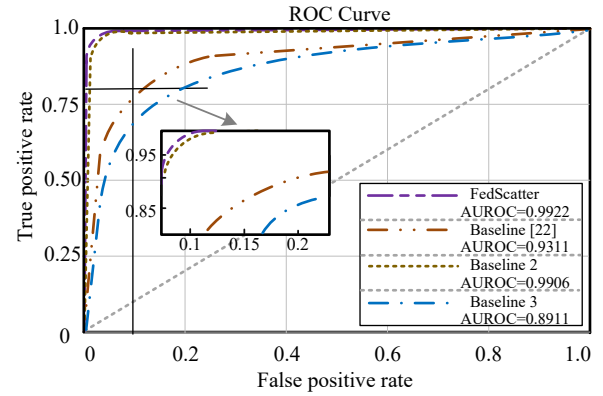


Fig. 10. ROC curve and AUROC of FedScatter and three baseline schemes under semi-honest attack scenario.

attacks scenario. As the figures show, FedScatter and baseline scheme 2 all keep a high performance under the semi-honest attacker, the accuracy of FedScatter is 97.66%, and the TPR and FPR of FedScatter are, respectively, more than 95.00% and less than 1.00%. The baseline scheme [22] gains an accuracy of 94.25%, and the TPR and FPR of 94.25% and 4.05%, respectively. The baseline scheme 3 achieves the accuracy of 90.63%, TPR of 62.88%, and FPR of 2.40%. Compared to the baseline scheme [22], the FPR of FedScatter is declining by 83.21%. Compared to the baseline scheme 3, the TPR of the FedScatter is increasing by a ratio of 33.88%, which could be that the authentication model in the baseline scheme 3 judges the identity signatures of mobile devices as illegitimate devices. Additionally, we use cosine similarity of the identity signature of continued two-round communication from a wireless IoT device to trace the moving state of a mobile wireless IoT device in FedScatter. This method can detect whether the identity signature from a wireless IoT device changes suddenly and then distinguish which devices are the semi-honest attackers. Based on Fig. 10, we can significantly observe that the cosine similarity we used in FedScatter can boost defense against semi-honest attackers. The AUROC of the FedScatter scheme improves by a ratio of 6.11% and a ratio of 11.35% compared to the baseline scheme [22] and the baseline scheme 3.

The performance of FedScatter and the baseline schemes under the powerful attacks scenario is shown in Fig. 11. We observe that the baseline scheme [22] and the baseline scheme

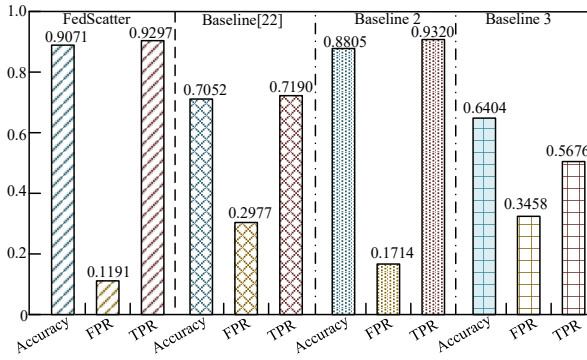


Fig. 11. Performance of FedScatter and three baseline schemes under powerful attacks scenario.

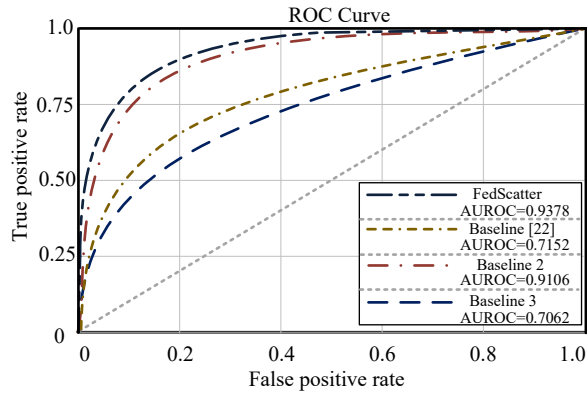


Fig. 12. ROC curve and AUROC of FedScatter and the three baseline schemes under powerful attacks scenario.

3 cannot resist the powerful attack at all with an accuracy of 72.52%, FPR of 29.77%, and TPR of 71.90% and an accuracy of 64.04%, FPR of 34.58%, and TPR of 56.76%, respectively. The FedScatter scheme and baseline 2 (FedAvg-based FL) scheme perform better than the baseline scheme [22] and the baseline scheme 3. FedScatter achieves high performance with an accuracy of 90.39% and a TPR of 90.71%, and the FedAvg-based baseline scheme has an accuracy of 88.05% and a TPR of 93.20%. The FPR of FedScatter is 11.91%, which is 30.51% lower than the FedAvg-based baseline scheme with FPR of 17.14%. Fig. 12 shows the ROC curve and AUROC of the FedScatter scheme and three baseline scheme. We can observe that FedScatter has an AUROC value of 0.9378, higher than the baseline scheme [22] with an AUROC value of 0.7152, the baseline scheme 2 with an AUROC value of 0.9106, and the baseline scheme 3 with an AUROC value of 0.7062, which proves that the performance of FedScatter is the best. However, the results are lower than those of the normal scenario and the semi-honest attack scenario. The main reason is that Algorithm 1 can eliminate the effects of injection of large samples from powerful attackers but can only alleviate the effects of the incorrect label. Since the ESs in a wireless IoT cluster for training the local model are trusted in FedScatter, the powerful attackers are hard to modify the signatures label even if they can gain prior knowledge about the construction of identity signatures, as analysis in Section V.

To conclude, FedScatter can maintain robustness and accuracy for the authentication of wireless IoTs under various attacks. Moreover, the backscatter-based signal propagation also achieves an effective identity signature for wireless IoTs in a lightweight way.

VII. CONCLUSION

This article presents FedScatter, a lightweight cross-domain authentication scheme for securing wireless IoTs. The FedScatter exploits the propagation feature of several lightweight backscatter tags fixed on the IoT to construct a “passive” identity signature, which can reduce hardware and arithmetic resource requirements for implementing static and mobile HW-IoT authentication. Besides, we use FL and a customized parameters aggregation algorithm to realize a secure cross-domain share of HW-IoTs’ identity signature knowledge and improve the authentication performance for the resource-constrained IoT clusters. We formally analyzed FedScatter’s security and implemented it with several backscatter tags, USRPs, and a server. We evaluate the performance of FedScatter under normal scenarios and various attack scenarios. The results show that FedScatter can achieve desirable performance and improve the security and robustness of HW-IoT authentication. In the future, we will continue our work by using backscatter to deploy practical wireless IoTs, such as wireless helmets and ad hoc robots, to realize a lightweight authentication for practical wireless IoT cross-domain collaboration. Moreover, we also focus on the backscatter tag’s hardware security, which can further improve the backscatter system’s reliability and credibility.

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